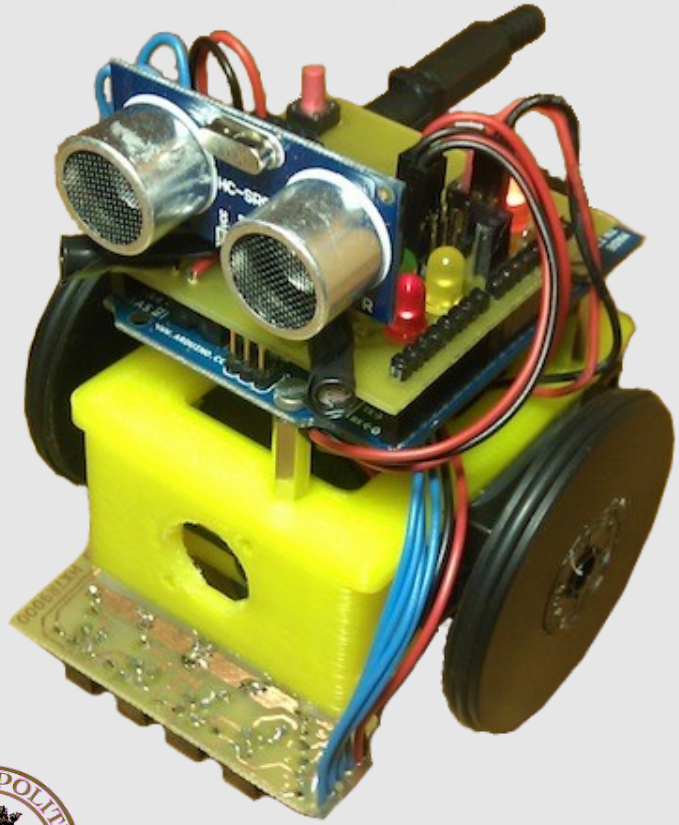
LOW COST EDUCATIONAL PLATFORM FOR ROBOTICS, USING OPEN-SOURCE 3D PRINTERS AND OPEN-SOURCE HARDWARE

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UAM UNIVERSIDAD AUTONOMA DE MADRID

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2012/11/19

ICERI 2012
5th International Conference of
Education, Research and Innovation

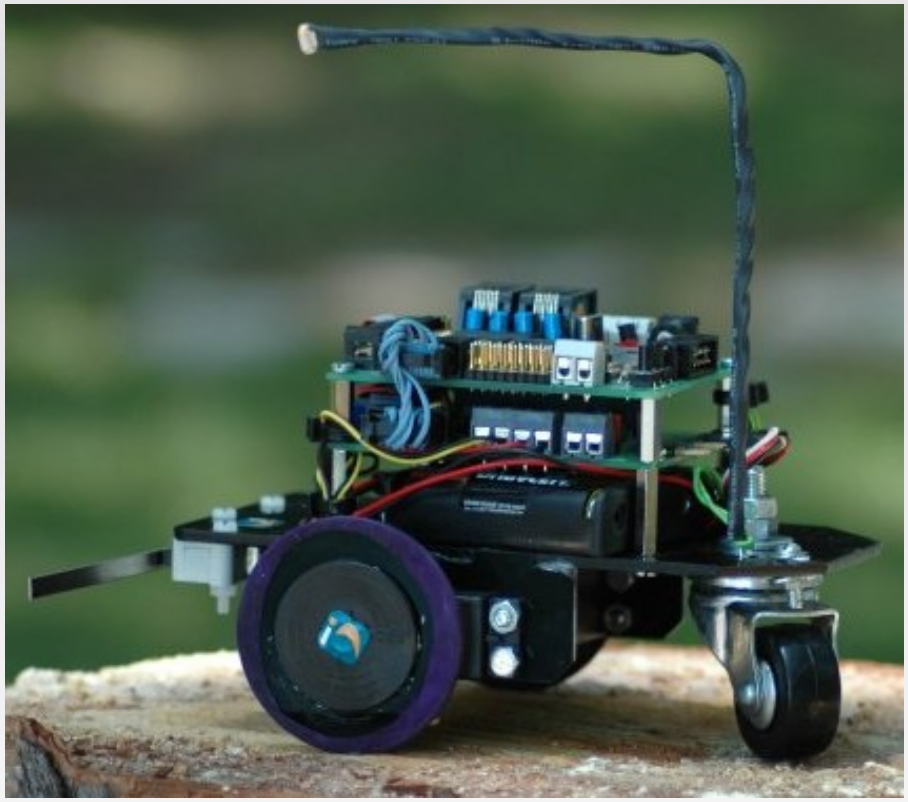
Robots in education

Interdisciplinary field:

- **Mechanics**
- **Electronics**
- **Programming**

Platform requirements:

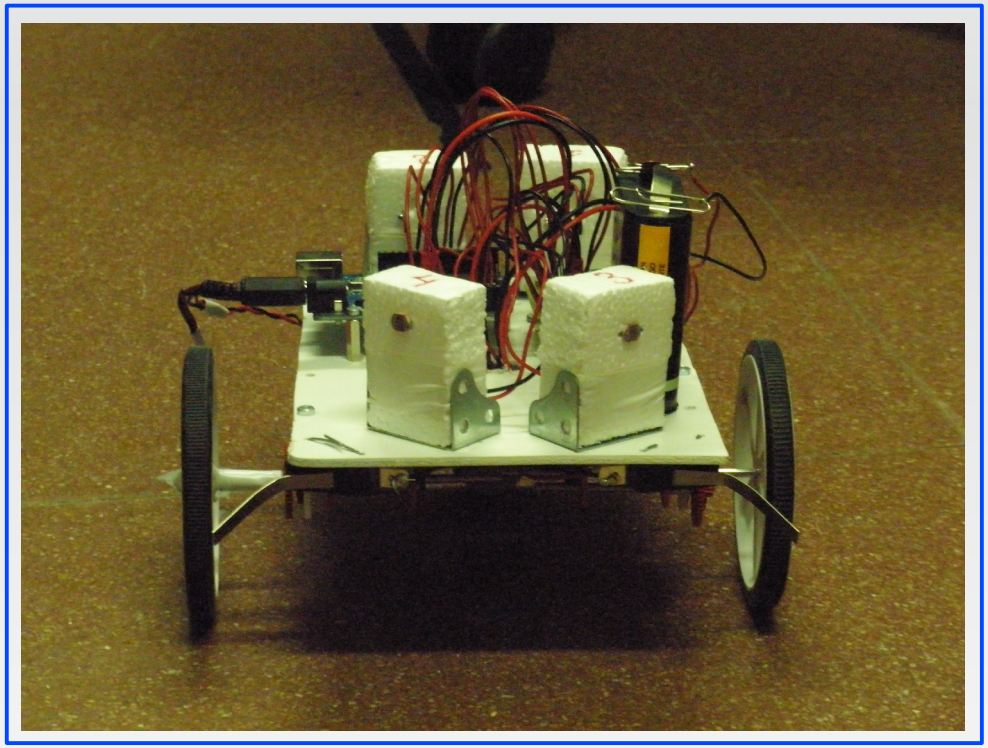
- **Low cost**
- **Easy to repair**
- **Flexible and expandable**



Type of platform

Commercial vs Custom



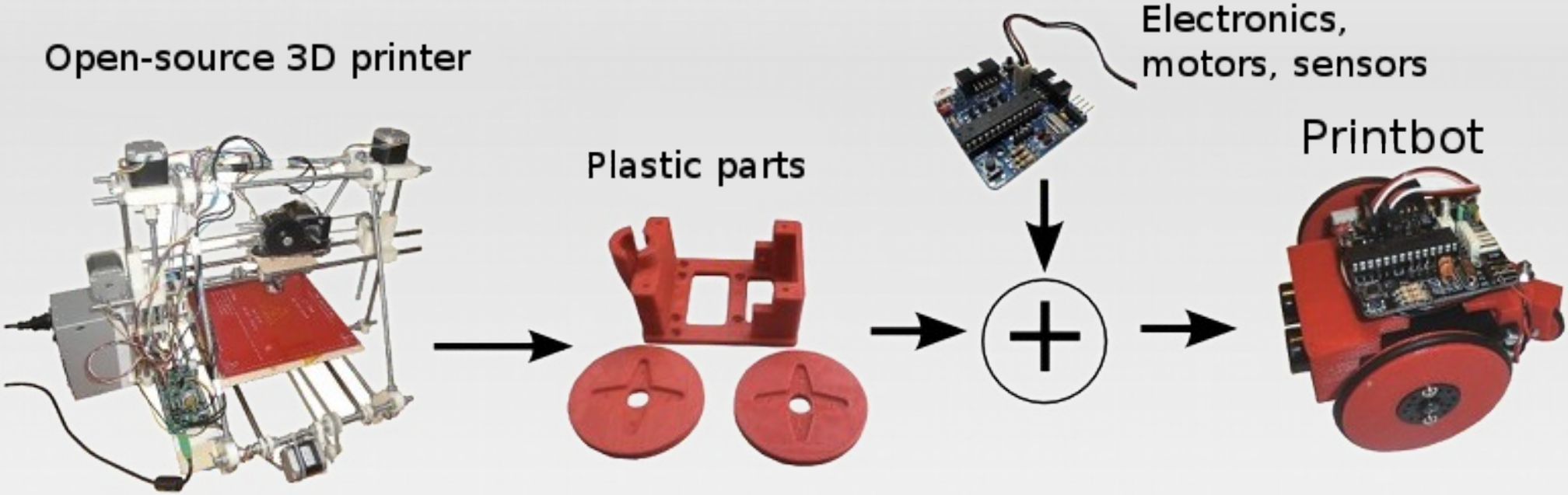


Low cost? Easy to repair? Flexible and expandable?

Industrial? **Home-made?**

Printbots

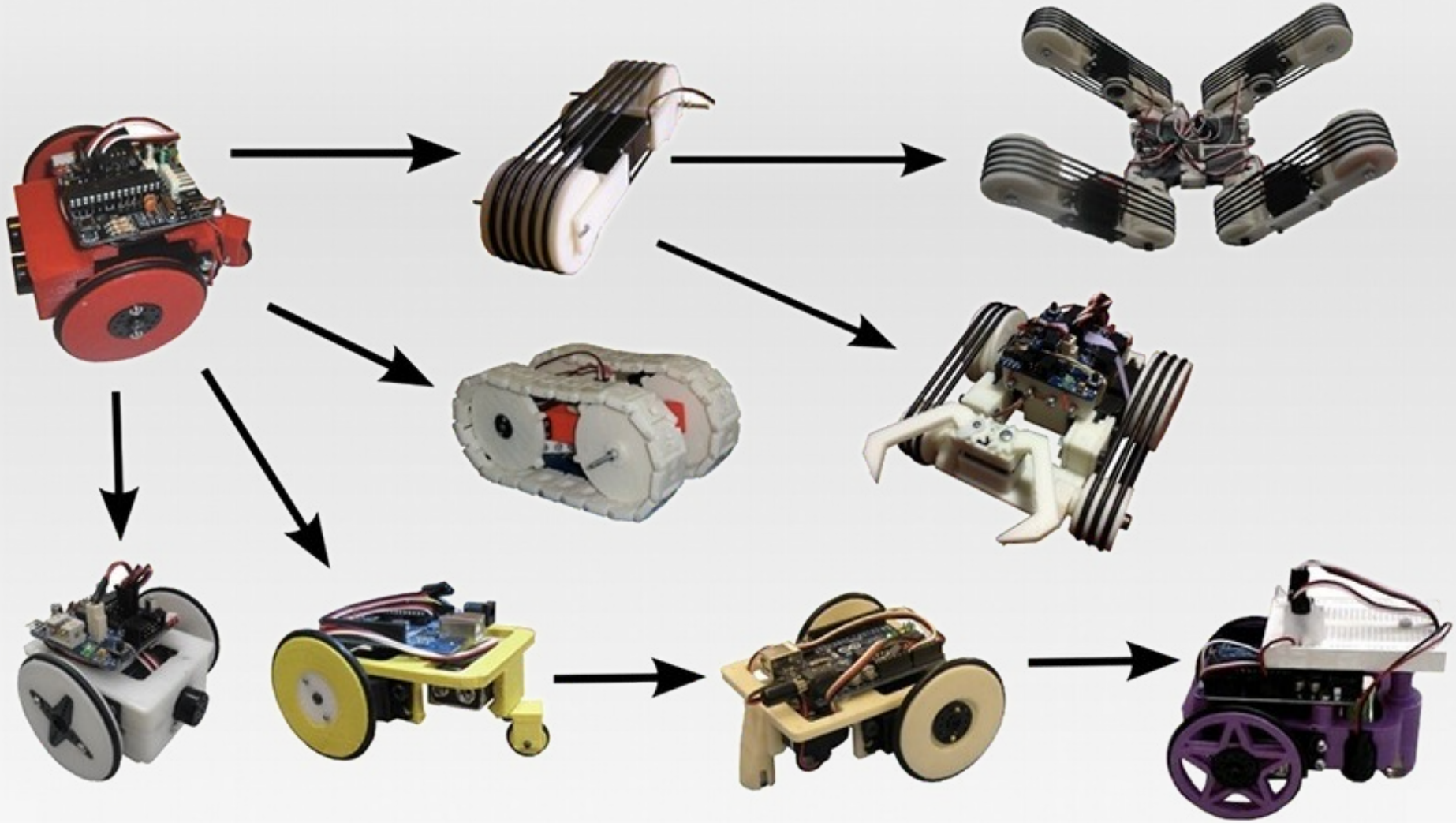
Build process:



- Custom
- As simple as required
- Mass producible

- **Open-Source design**
- **Shared by the community**
  www.thingiverse.com... etc

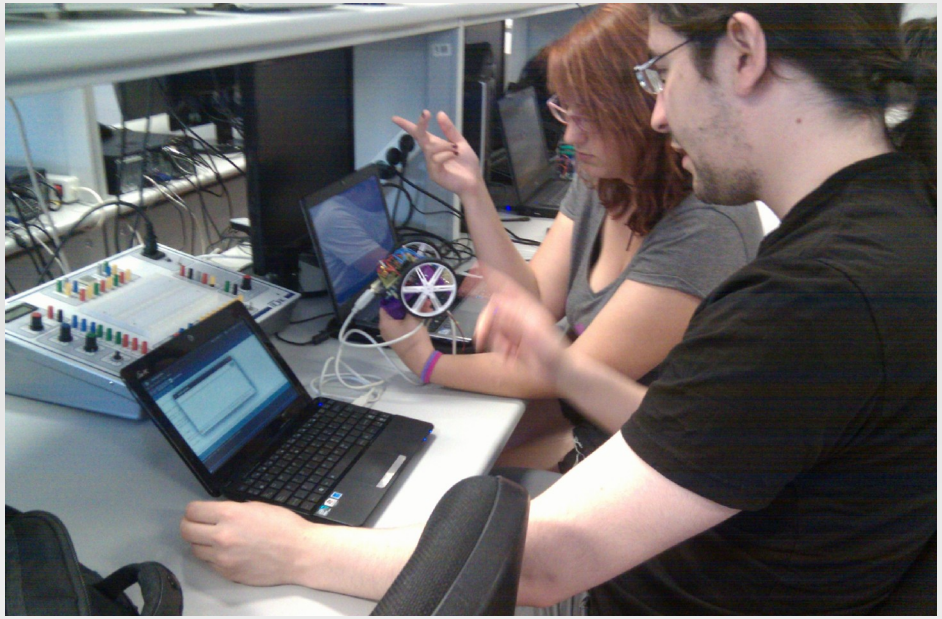
Evolution and diversification



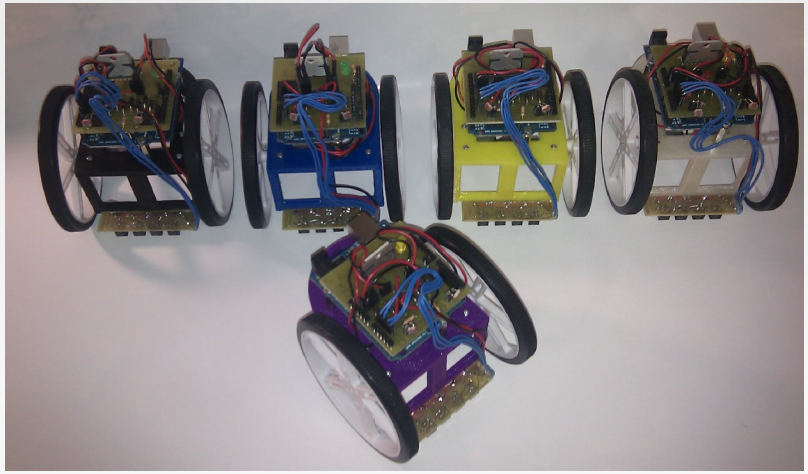
Fast growing open-hardware ecosystem!

Robotics introductory course

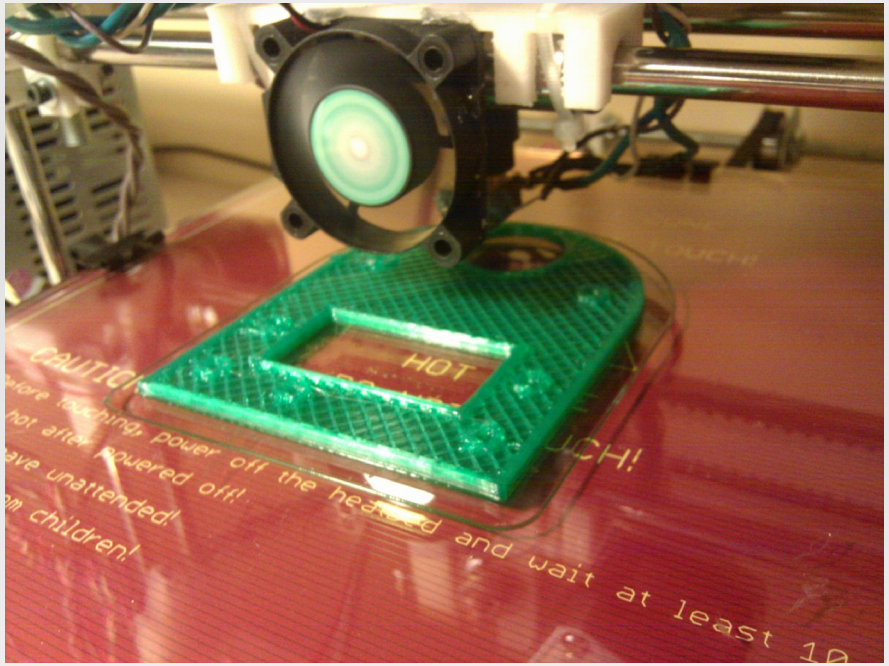


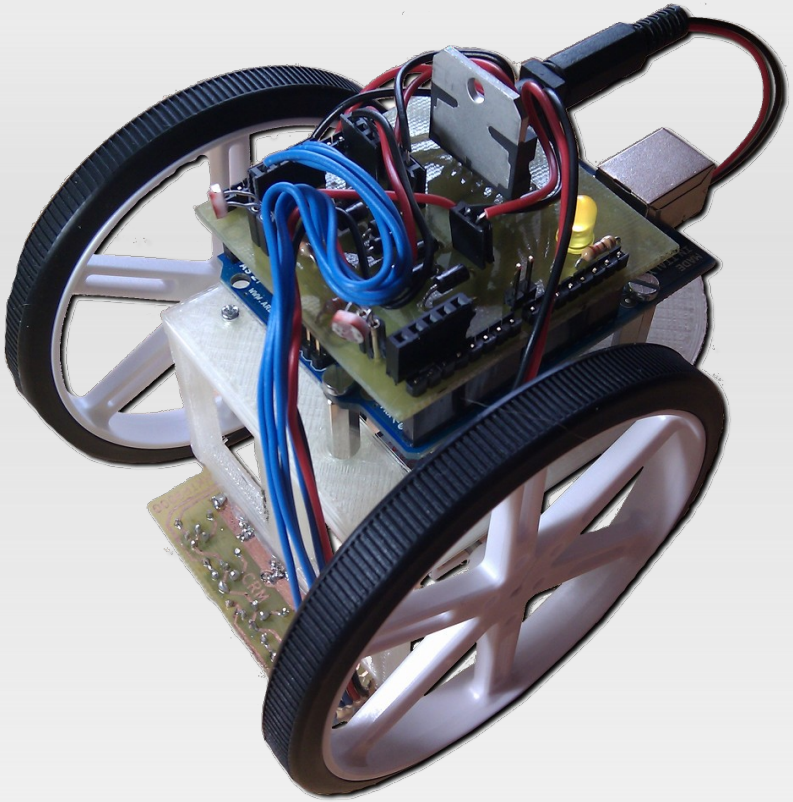


- **20 engineering students**
- **15 hours (5 days)**
- **Robotics and Mechatronics group (UAM Madrid)**



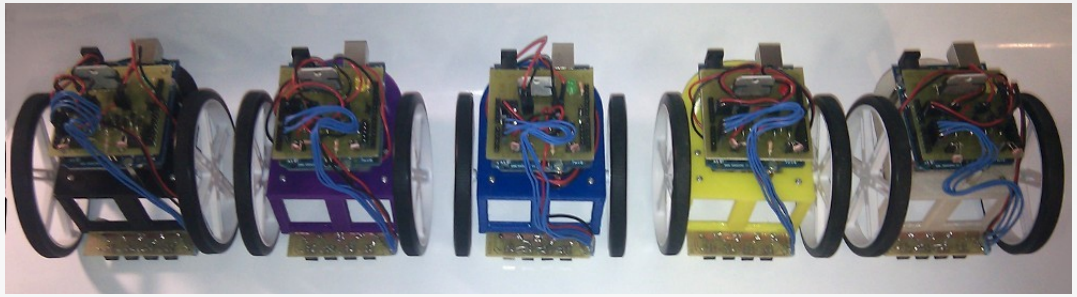
Our first design



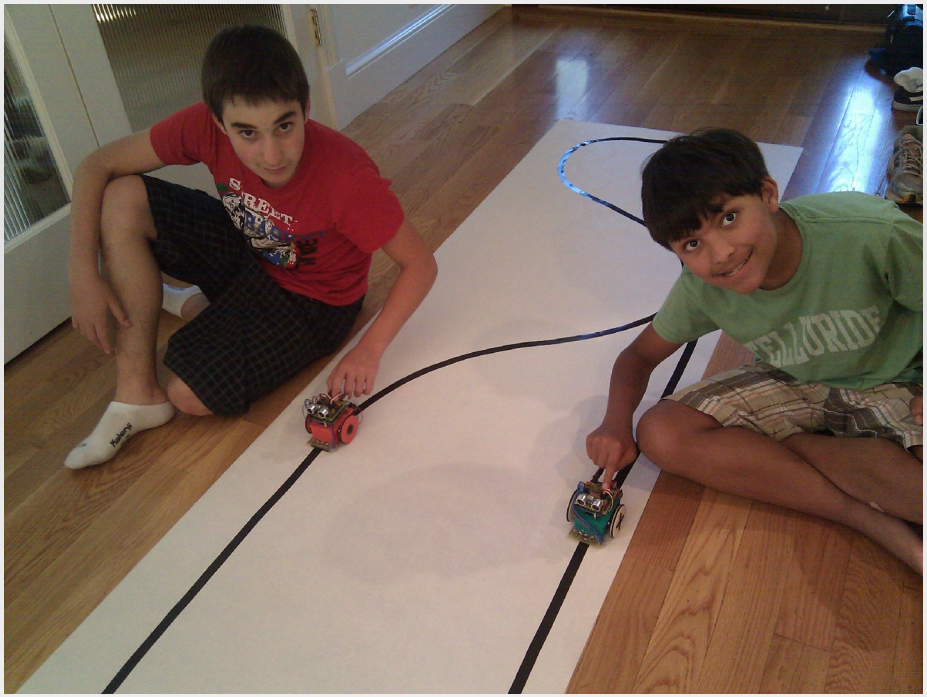


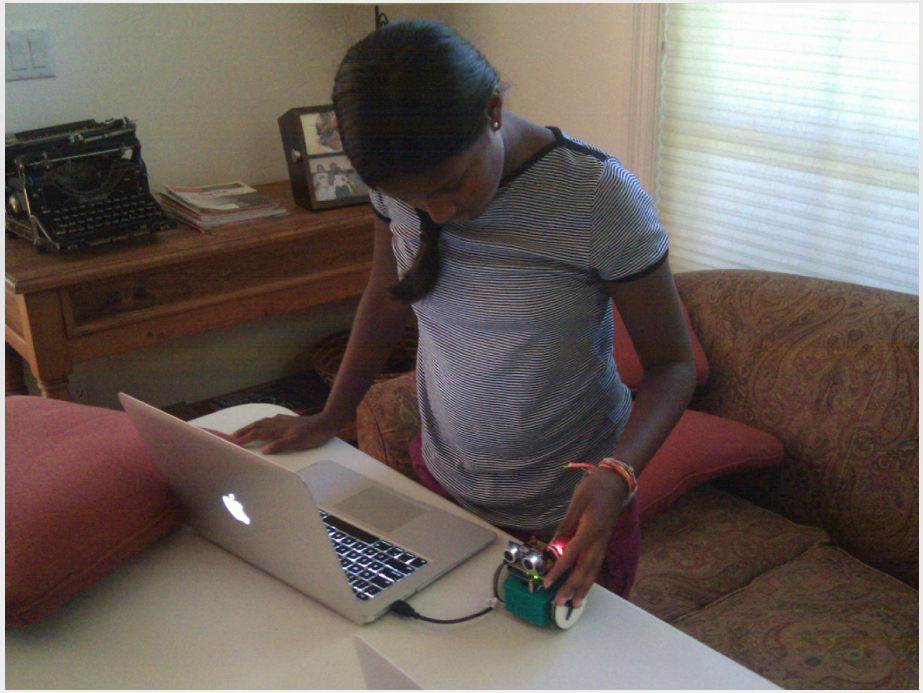
Available resources:

- **Arduino boards**
- **Motors and wheels**
- **Simple PCB etching**

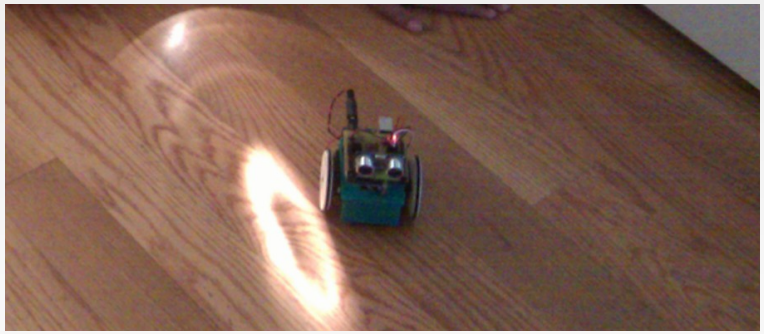


ArduSkybot Summer Workshop





- **4 primary students**
- **10 hours (4 days)**
- **Palo Alto, California (USA)**

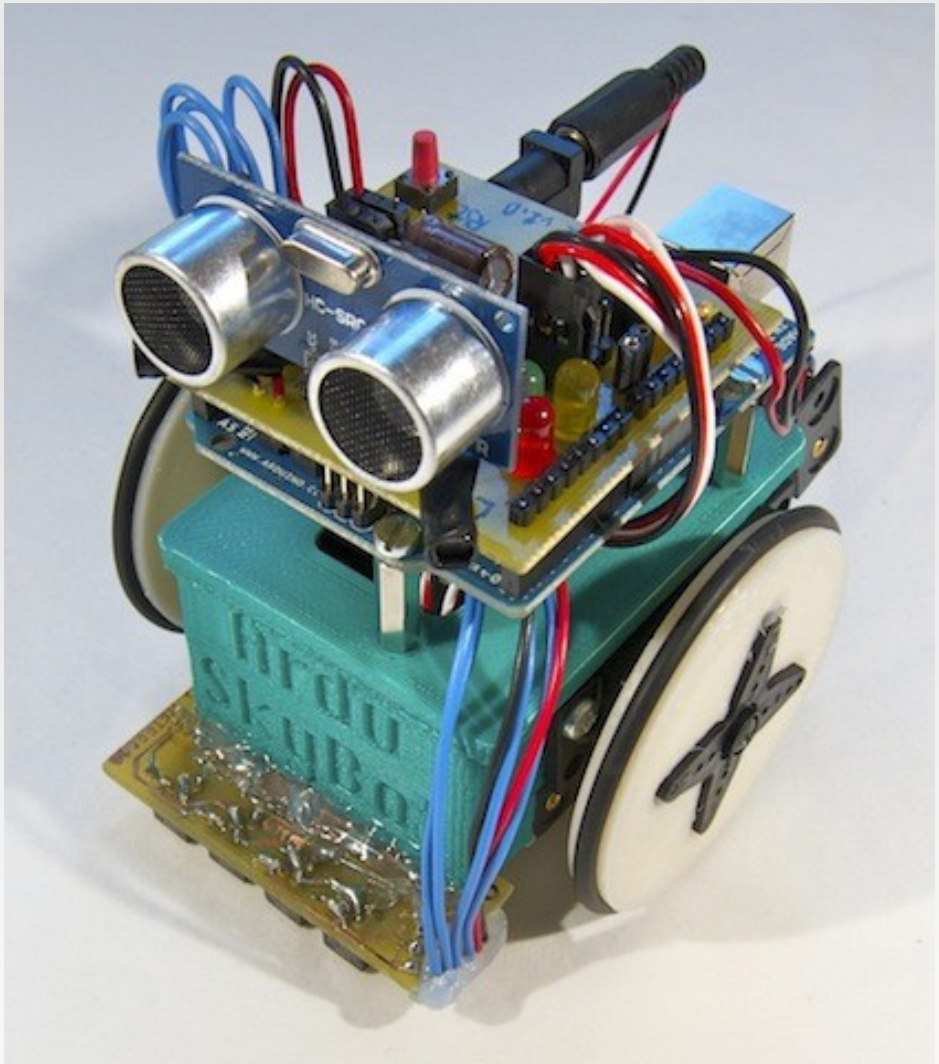


The ArduSkyBot

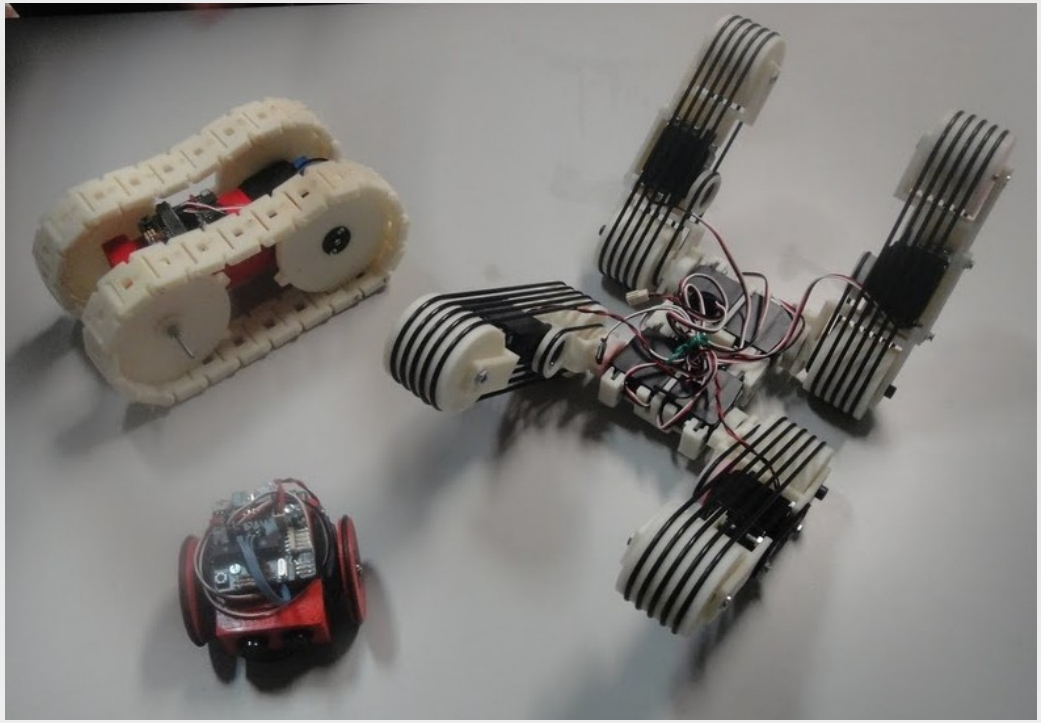
- **Various sensors/actuators**
- **Simple assembly**

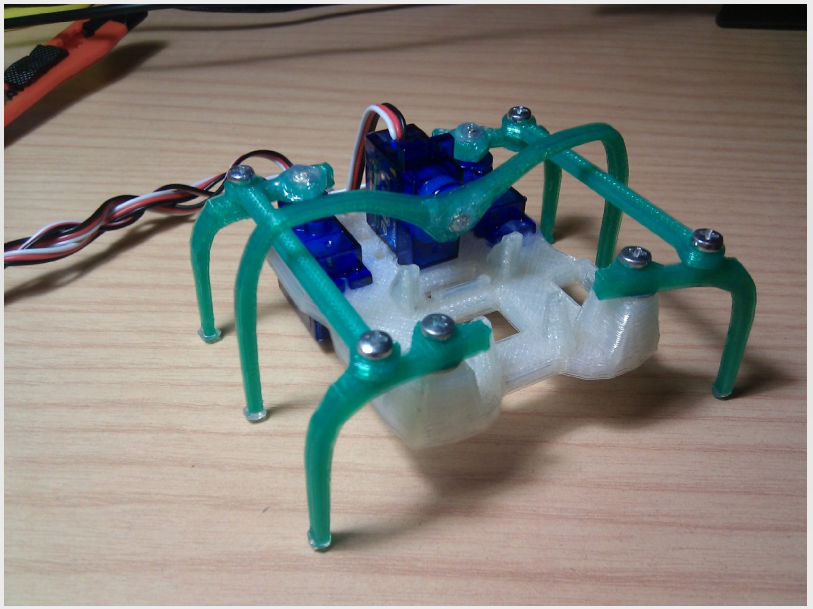
- **Based on the MiniSkybot and the HKTR-9000**

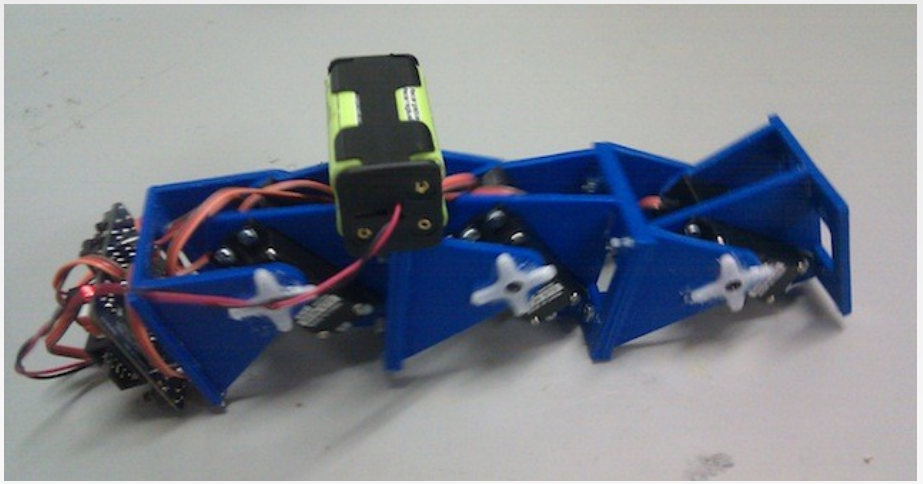
- **Designed in one week**
- **Four robots ready in record time!**



Printbots are great!





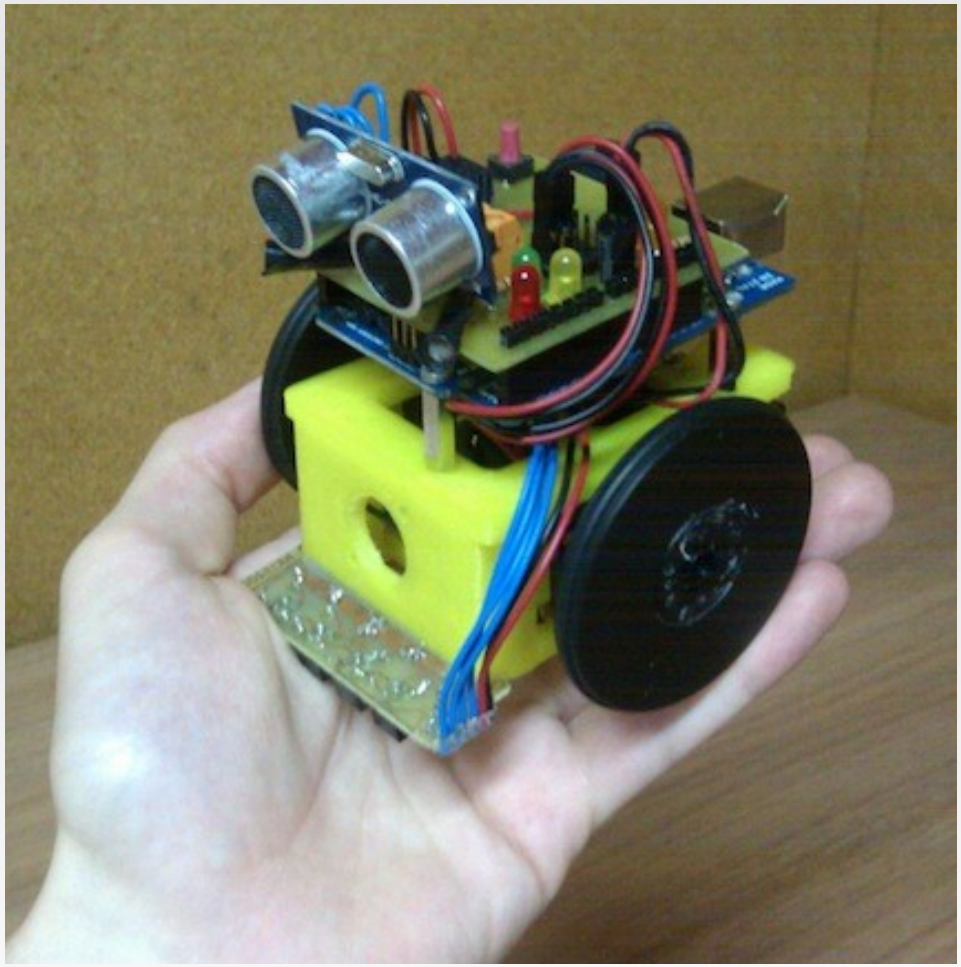


This design approach will work with any kind of robotics course

Conclusions/Discussion

Using printbots simplifies setting up robotic courses:

- **Small setup effort**
- **Custom platform**
- **Fast robot replication**
- **Low cost**
- **Use of available resources**
- **New way of technological education**



Questions are welcome

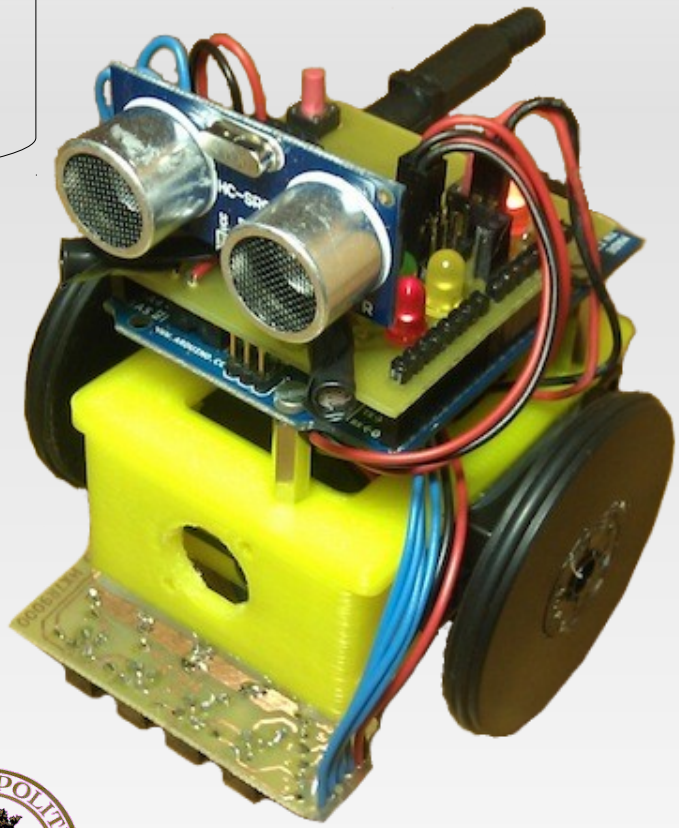
Thanks for your attention!

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Dr. Juan González-Gómez[2]

More information:

[1] http://www.carlosgs.es

[2] http://www.iearobotics.com



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